

Homework 17

Please review everything we have studied so far from Chapter 5 (most of 5.1 and 5.2, skipping 5.2.1).

Focus on the following:

- General idea of density function (for a continuous random variable) and distribution function (for any random variable). How to calculate distribution function from density function (integrate) and density function from distribution function (differentiate).
- Formulas for density functions, distribution functions, expected values, and variances for exponential and continuous uniform distributions.

Those points need to be on your note sheet for Midterm 2.

1. In Homework 15, you thought about the chance experiment of picking two numbers independently from the uniform distribution on $[0, 10]$. Let Y be the random variable giving the higher of the two numbers. In Homework 15, I provided the density function for the continuous random variable Y . Let's see how I figured it out.

Start by calculating the distribution F function for Y . Let x be a real number. You will calculate $F(x) = P(Y \leq x)$ in the following steps.

- a. Suppose that $x < 0$. What is $F(x)$?

$$F(x) = 0 \text{ if } x < 0.$$

- b. Suppose that $x > 10$. What is $F(x)$?

$$F(x) = 1 \text{ if } x > 10.$$

- c. Let X_1 be the random variable representing the first number picked and X_2 be the random variable representing the second number picked. The random variable Y is the maximum of X_1 and X_2 . The event $(Y \leq x)$ is thus $(X_1 \leq x) \cap (X_2 \leq x)$. Suppose $0 \leq x \leq 10$. Calculate $P((X_1 \leq x) \cap (X_2 \leq x))$ using the independence of X_1 and X_2 as well as the formula for the distribution function for a uniformly distributed continuous random variable (Example 5.7).

$$P(Y \leq x) = P((X_1 \leq x) \cap (X_2 \leq x)) = P(X_1 \leq x)P(X_2 \leq x) = \left(\frac{x}{10}\right)\left(\frac{x}{10}\right) = \frac{x^2}{100}$$

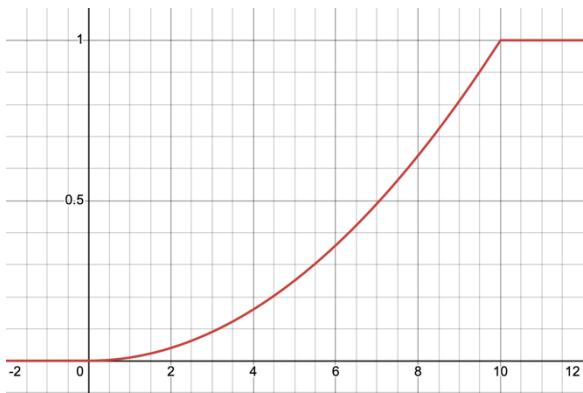
- d. Combine the three previous parts to write a complete formula for $F(x)$, valid for any real number x .

$$\text{If } x < 0, F(x) = 0. \text{ If } x > 10, F(x) = 1. \text{ If } 0 \leq x \leq 10, F(x) = \frac{x^2}{100}.$$

- e. Sketch the graph of this distribution function.

The following Desmos link shows the distribution function:

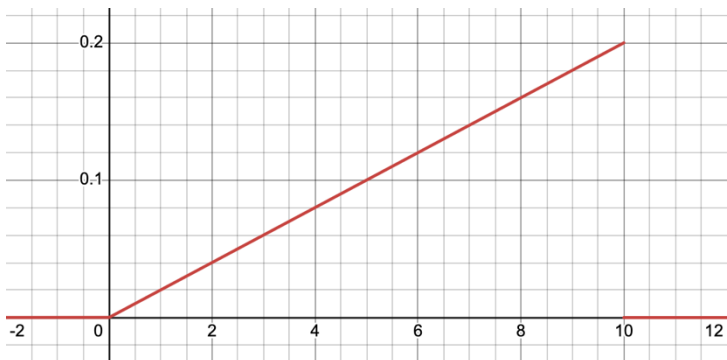
<https://www.desmos.com/calculator/yx6tyiomkr>.



f. Calculate the density function f for Y .

The density function f is the derivative of the distribution function. For $x < 0$, $f(x) = 0$, for $0 \leq x \leq 10$, $f(x) = \frac{x}{50}$, and for $x > 10$, $f(x) = 0$. This density function is discontinuous at 10. We could use any value for $f(10)$: $f(10) = 0.2$ (as above), $f(10) = 0$, or anything else. When we integrate f to calculate probabilities, values at individual points do not affect the integrals.

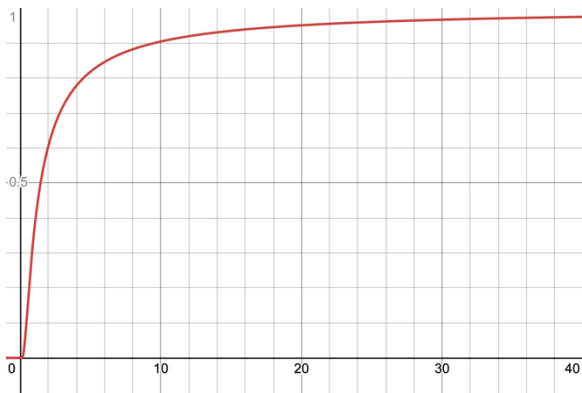
Here is a graph of the density function: <https://www.desmos.com/calculator/s6pyfepg3>.



2. Consider the function F defined by $F(x) = 0$, if $x \leq 0$, and $F(x) = e^{-\frac{1}{x}}$, if $x > 0$. This function is a continuous density function in the sense that its limiting value toward $-\infty$ is 0, its limiting value toward $+\infty$ is 1, and it is non-decreasing.
 - a. Use a graphing tool such as Desmos (<https://www.desmos.com>) or a graphing calculator to graph F . Adjust your viewing window to highlight the interest parts of the graph. Focus on positive values of x , since $F(x) = 0$ if $x < 0$. The outputs of F are between 0 and 1, and so you only need this range on the y-axis. Explore the graph to find a suitable range of input values.

Sketch your graph below, including scales on the axes.

Here is a Desmos graph: <https://www.desmos.com/calculator/fs4lw6z0xh>.

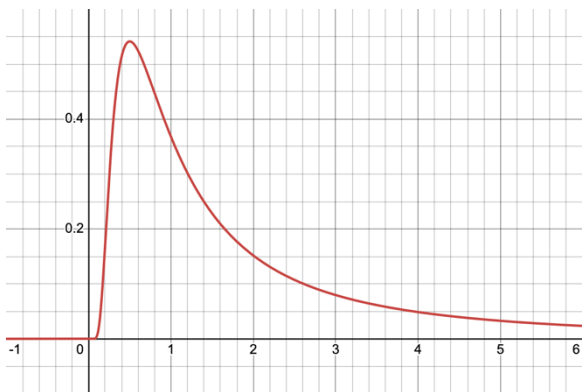


- b. Calculate the density function f for this distribution. Your formula for $f(x)$ should work for any input value x .

Using the chain rule, we find that for $x \leq 0$, $f(x) = 0$ and for $x > 0$, $f(x) = e^{-\frac{1}{x}} \cdot \frac{1}{x^2}$.

- c. Graph the density function with a graphing tool. Sketch your graph below including scales on the axes. Adjust your viewing window to highlight the interesting parts of the graph. Focus on positive x -values, since $f(x) = 0$ for $x < 0$.

Here is a Desmos graph: <https://www.desmos.com/calculator/qdvn3oclkr>.



- d. What value appears to be the “mode” for the distribution? The mode is the input x where the density $f(x)$ has its maximum value. You could solve for x using calculus by finding critical points of f . I am just asking you to estimate using your graph of f .

The mode of the density function appears to be at input 0.5.

I didn't ask you to do it, but we can confirm the location of the mode using calculus as follows: for $x > 0$, $f'(x) = e^{-\frac{1}{x}} \cdot \left(\frac{1}{x^4}\right) + e^{-\frac{1}{x}} \cdot (-2) \cdot \left(\frac{1}{x^3}\right)$. To find the critical points of f , we solve $f'(x) = 0$ (looking at $x > 0$).

$$\begin{aligned}
 0 &= f'(x) \\
 0 &= e^{-\frac{1}{x}} \left(\frac{1}{x^4}\right) + e^{-\frac{1}{x}} (-2) \left(\frac{1}{x^3}\right) \\
 2e^{-\frac{1}{x}} \left(\frac{1}{x^3}\right) &= e^{-\frac{1}{x}} \left(\frac{1}{x^4}\right) \\
 2 \left(\frac{1}{x^3}\right) &= \frac{1}{x^4} \text{ since } e^{-\frac{1}{x}} \text{ is non-zero}
 \end{aligned}$$

$$2x = 1$$

$$x = \frac{1}{2}.$$

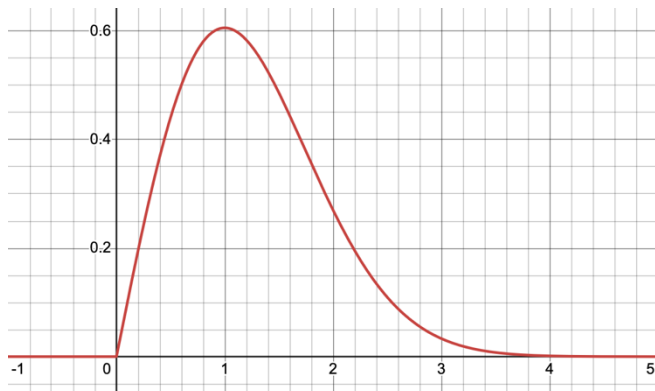
Therefore, the only critical point is at input $\frac{1}{2}$. Judging from the graph, we see that this input gives the largest output. (We could use more Calculus 1 techniques—the second-derivative test—to analyze the situation without relying on the graph.)

3. A continuous probability distribution used for scientific purposes has the following density function f :

$$f(x) = 0, \text{ if } x < 0, \text{ and } f(x) = xe^{-\frac{x^2}{2}}, \text{ if } x \geq 0.$$

- a. Use a graphing calculator or Desmos to graph f . Adjust your viewing window to see the important features of the graph. Focus on positive values of x , since $f(x) = 0$ if $x < 0$. Look at y -values between 0 and 1. Sketch the graph of f , including scales on the axes.

Here is a Desmos link with the graph: <https://www.desmos.com/calculator/utstv2kuto>.



- b. Use your graph to estimate the mode of the density function f . The mode is the input x where the density $f(x)$ has its maximum value.

The mode appears to be at the input 1.

We could check using calculus. To find the critical points of f on $(0, \infty)$, we solve $f'(x) = 0$ where x is a positive real number. In this range of input values, we have $f'(x) = e^{-\frac{x^2}{2}} + xe^{-\frac{x^2}{2}}(-x)$. Solving for the critical points with $x > 0$, we find

$$0 = e^{-\frac{x^2}{2}} + xe^{-\frac{x^2}{2}}(-x)$$

$$x^2 e^{-\frac{x^2}{2}} = e^{-\frac{x^2}{2}}$$

$$x^2 = 1 \text{ since } e^{-\frac{x^2}{2}} \text{ is non-zero}$$

$$x = 1 \text{ since } x > 0.$$

- c. Find an antiderivative of $xe^{-\frac{x^2}{2}}$, i.e. calculate $\int xe^{-\frac{x^2}{2}} dx$. Use the substitution technique with $u = \frac{x^2}{2}$.

We use the substitution $u = \frac{x^2}{2}$, which gives us $du = xdx$.

$$\begin{aligned}\int x e^{-\frac{x^2}{2}} dx &= \int e^{-u} du \\ &= -e^{-u} \\ &= -e^{-\frac{x^2}{2}}\end{aligned}$$

- d. Let X be a continuous random variable having the distribution described by the density function f . Calculate $P(1 \leq X \leq 3)$.

$$\begin{aligned}P(1 \leq X \leq 3) &= \int_1^3 x e^{-\frac{x^2}{2}} dx \\ &= \left[-e^{-\frac{x^2}{2}} \right]_{x=1}^{x=3} \\ &= e^{-\frac{1}{2}} - e^{-\frac{9}{2}} \\ &\approx 0.5954\end{aligned}$$

- e. Find the *median* of X . This means finding m such that $\frac{1}{2} = P(X \leq m) = \int_0^m f(x)dx$. You can do this! Calculate $\int_0^m f(x)dx$ using your antiderivative. Then set this function of m equal to $\frac{1}{2}$, and solve for m .

We solve $\frac{1}{2} = P(X \leq m)$ as follows: The equation we need to solve is the following:

$$\begin{aligned}\frac{1}{2} &= P(X \leq m) \\ &= \int_0^m f(x)dx \\ &= \left[-e^{-\frac{x^2}{2}} \right]_{x=0}^{x=m} \\ &= 1 - e^{-\frac{m^2}{2}}.\end{aligned}$$

Now we solve $\frac{1}{2} = 1 - e^{-\frac{m^2}{2}}$:

$$\begin{aligned}\frac{1}{2} &= 1 - e^{-\frac{m^2}{2}} \\ e^{-\frac{m^2}{2}} &= \frac{1}{2} \\ -\frac{m^2}{2} &= \ln\left(\frac{1}{2}\right) \\ m^2 &= -2 \ln\left(\frac{1}{2}\right) \\ m &= \sqrt{-2 \ln\left(\frac{1}{2}\right)} \text{ since } m > 0 \\ &\approx 1.1774\end{aligned}$$